

Course Outcomes:	
On completion of the course students will be able to	
PCC- CS502.1	Understand basic concepts of operating system and process management.
PCC- CS502.2	Analyse various scheduling algorithms and process synchronization.
PCC- CS502.3	Evaluate various deadlock prevention and avoidance algorithms.
PCC- CS502.4	Design various memory, file and disk management schemes.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	1	2	1	1	1	1	1	1	-	2	2	1	2
CO2	3	3	3	3	2	-	1	-	2	2	2	2	3	3	3
CO3	2	2	2	3	2	-	-	-	2	2	1	2	2	2	3
CO4	2	3	2	3	2	-	1	1	3	2	2	3	2	3	3
AVG.	2.25	2.25	2.00	2.75	1.75	0.25	0.75	0.50	2.00	1.75	1.25	2.25	2.25	2.25	2.75

Machine Learning

Code: PCC-CS503

Contact: 3L

Theory: 3 hrs. / Week

Credit Points: 3

Course Objective:	
1.	To learn the concept of how to learn patterns and concepts from data without being explicitly programmed
2.	To design and analyses various machine learning algorithms and techniques with a modern outlook focusing on recent advances.
3.	Explore supervised and unsupervised learning paradigms of machine learning.
4.	To explore Deep learning technique and various feature extraction strategies.

Pre-Requisite:	
1.	Basic understanding of Linear algebra, Trigonometry, Statistics, calculus, probability
2.	Python Programming

Unit	Content	Hrs.
1.	Some performance metrics associated with ML Supervised Learning (Regression/Classification) Basic methods: Distance-based methods, Nearest-Neighbor's, Decision Trees, Naive Bayes Linear models: Linear Regression, Logistic Regression Support Vector Machines, Nonlinearity and Kernel Methods	12
2.	Unsupervised Learning Clustering: K-means/Kernel K-means / k-medoids Dimensionality Reduction: PCA and kernel PCA Matrix Factorization and Matrix Completion	9

3.	Evaluating Machine Learning algorithms and Model Selection, Introduction to Statistical Learning Theory, Ensemble Methods (Boosting, Bagging, Random Forests)	6
4.	Time-Series Data, basics of Semi-supervised Learning, Reinforcement Learning, DL fundamentals: Introduction to neural network and feed forward network	9

References:

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer 2009 (freely available online)
2. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007
3. Dr. Rajiv Chopra, Machine Learning, Khanna Publishing House, 2018
4. Tom M. Mitchell, Machine Learning, McGraw-Hill, 2017
5. Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012

Course Outcomes:	
On completion of the course students will be able to	
PCC- CS503.1	To understand principles and concepts of various ML algorithms
PCC- CS503.2	To evaluate the ML algorithms using real life data
PCC- CS503.3	To analyse the ML algorithms according to different performance metrics
PCC- CS503.4	To understand Ensemble Methods, time series data, semi supervised learning, Reinforcement Learning, and essentials of NN architectures

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	-	-	-	-	-	-	-	3	3	3	1
CO2	3	2	2	2	1	-	-	-	-	1	-	3	3	3	2
CO3	3	3	2	3	1	-	-	-	-	1	-	3	3	3	3
CO4	2	3	3	3	-	-	-	-	1	2	-	3	3	3	3
AVG.	2.83	2.83	2.17	2.50	1.00	1.00	1.00	0.00	1.00	1.33	0	3.00	3.00	3.00	2.50

Introduction to Industrial Management (Humanities III)

Code: HSMC 501

Contacts: 3L

Theory: 3 hrs. / Week

Credit Points: 3

Course Objective:	
1.	Gain a comprehensive understanding of industrial systems, management principles, and key concepts such as production planning, quality management, and supply chain operations.
2.	Develop practical skills in operations management, leadership, decision-making, and communication to effectively manage industrial processes, teams, and resources.
3.	Stay updated on emerging trends, technologies, and best practices in industrial management to adapt, innovate, and lead change initiatives in dynamic industrial environments.